Occupational Health and Safety-1

Psychosocial risk factors
Ergonomics

Psychosocial risk factors

WORK ASSOCIATED PSYCHOSOCIAL RISK FACTORS

1. Nature of the Job
2. Workload and Work Pace
3. Time shift
4. Control
5. Environment and Equipment
6. Corporate Culture
7. Interpersonal Relations
8. Job in the Workplace
9. Career Development
10. Work and Non-Work Life Interaction

Nature of the Job

- **Those who work in workplaces where mass production technology is dominant:**
  - narrow content,
  - inability to use skills
  - lack of job diversity
  - repetitive job,
  - uncertainty,
  - not having the opportunity to learn,

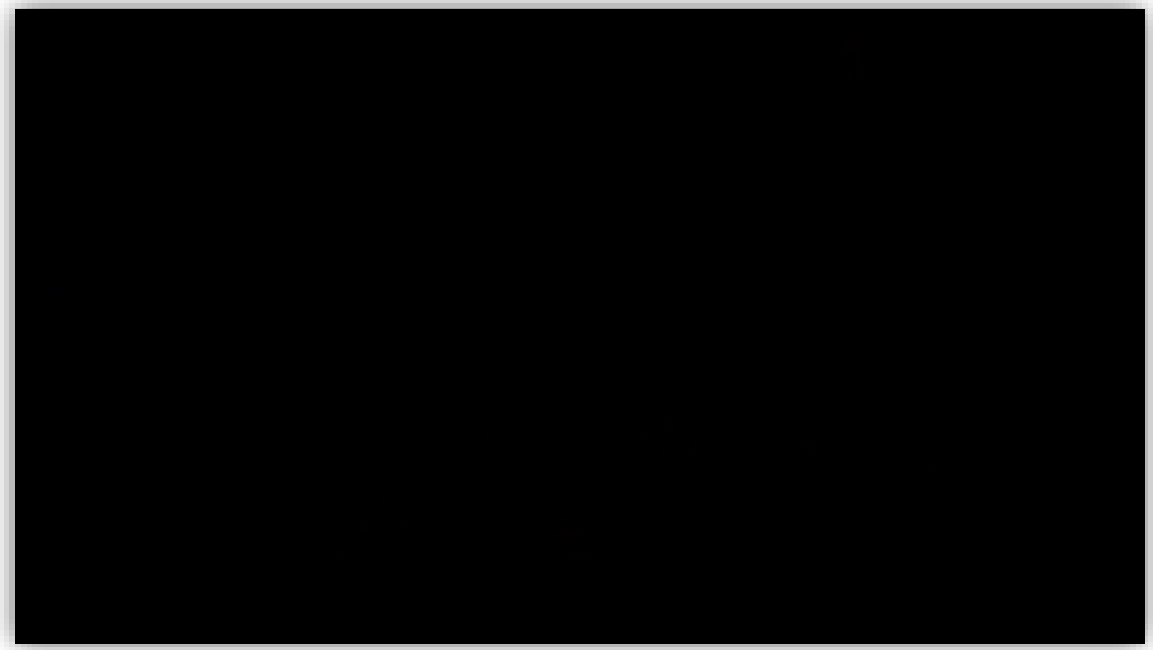


ABB Robotics - 10 most popular applications for robots

Nature of the Job

- **Feelings and behaviors that lead to inefficiency:**
  - loneliness,
  - alienation from work and alienation from each other,
  - excessive stress
  - monotony

- **Effects on the employee; advanced psychological and social disorders**
  - hopelessness
  - distress
  - tension
  - apathy
  - passive resistance
  - aggression

Workload and Pace

- ❑ Quantitative workload
- ❑ Qualitative workload

Workload and Pace

- **Qualitative workload**
  - There is a mismatch between the qualifications required by the job and the qualifications of the person
  - Accordingly, when the employee who will do the job does not have the knowledge, ability and personal characteristics required by the job, this job will be difficult for him.

Workload and Pace

- **Quantitative workload**
  - Quantitative workload can be explained as the lack of time for a particular job to be done by a person, the fact that there are many different jobs to be done at a particular time, and the work to be done is physically heavy and tiring.

Time Shift

- In the shift working order, the working hours in the workplace are spread over 24 hours and the working hours are changed periodically.

- Due to the shift work pattern, many sleep problems can be experienced in the employees, and accordingly, different physical and mental complications may occur.

Control

➢ Employee involvement in decision-making processes, not just decisions

Benefits:

- Job satisfaction increases
- Threats become opportunities
- Self-confidence grows
- Professional skills improve

Result: Better decisions through active participation and ownership

Environment and Equipment

- Employees are the most important elements of business systems. For this reason, systems, tools and equipment should be designed according to the employee's limits of thought and potential abilities.

Environment and Equipment

- With the improvement of the working environment, the morale and motivation of the employees will be high.

Corporate Culture

Corporate culture identifies and highlights the beliefs and behaviors of your organization. Overall, it guides how your employees should act, feel, and think.

The existence of an effective communication in the workplace is faced by the management levels as a basic necessity in achieving organizational goals and increasing productivity.

One of the most important factors in creating a corporate culture is education.

Interpersonal Relations

- A manager whose scientific and technical aspect predominates is viewed as more stressful than one who employs interactive management techniques.
- Consequently, the individual who works in constant fear and anxiety of being punished will experience stress.

- For an employee who is not appreciated by his superior and cannot find the type of relationship he is seeking in the workplace, the working environment can become extremely monotonous and unbearable.

Interpersonal Relations

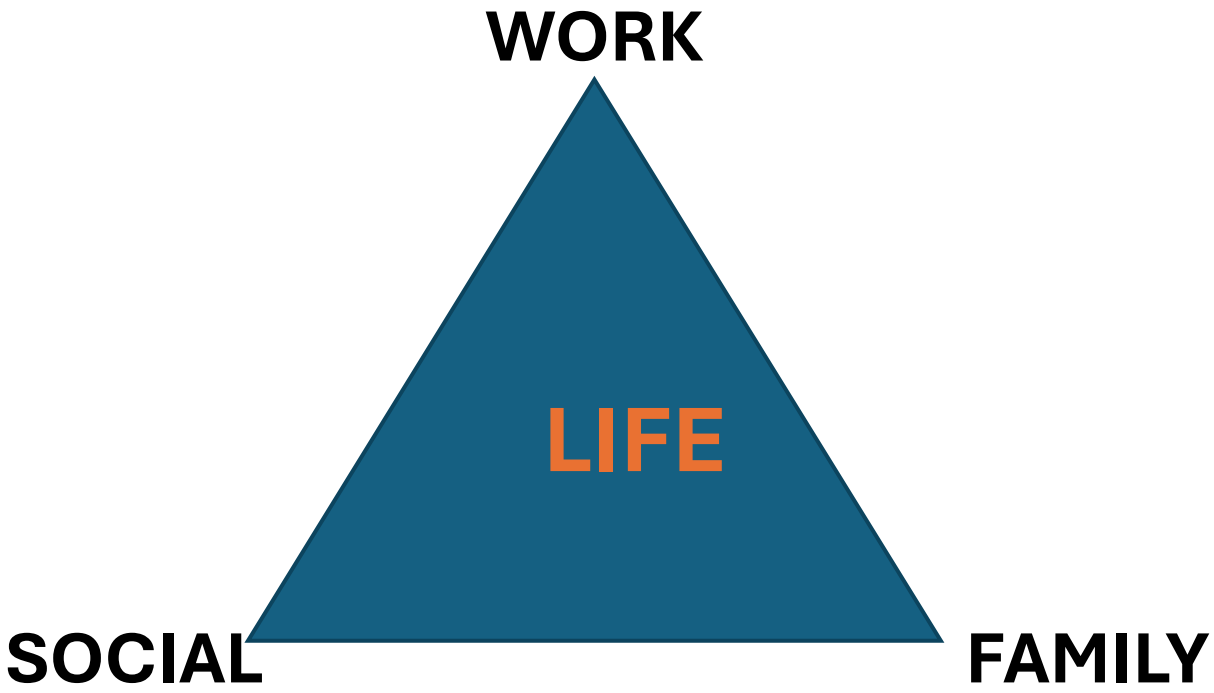
Task at Work

- The manager has inconsistent and contradictory expectations from the employee
- Multiple managers make requests to the same employee that involve overlapping role behaviors
- Role overloading of the employee beyond his/her capacity
- The employee does not know what he is authorized to do in his job

Career Development

- more power, prestige and money
- Since personnel who are unable to utilize their knowledge and skills effectively cannot offer their full capacity, they may experience tension due to being unable to rise and being blocked.

Work and Non-Work Life Interaction



Work and Non-Work Life Interaction

- Sources of stress that are not work-related but affect an individual at work
  - The birth of a new child
  - Low giving up a hobby
  - Unexpected spending
  - Low wages
  - not being able to live a regular life

Changes in the Individual During Stress

- Alert Phase
- Resistance Phase
- Exhaustion Phase

Changes in the Individual During Stress

Alert Phase:

- When the individual encounters a source of stress, the body displays a "fight or flight" response due to the activation of nervous system.
- As a result of the physical and chemical changes that occur in the body during the fight or flight response, the person becomes ready to face the source of stress or to escape.
- This situation develops in the form of an acceleration of the heartbeat, an increase in blood pressure, an acceleration of respiration and a sudden release of adrenaline.

Changes in the Individual During Stress

Resistance Phase:

Adaptation Stage: The body attempts to stabilize and cope with the stressor.

Physical Recovery: Heart rate, blood pressure, and breathing normalize; muscle tension decreases.

Energy Restoration: Efforts focus on regaining lost energy and repairing physical strain.

Behavioral Signs: Stressed behaviors persist as the individual continues resisting the stressor.

Time-Limited: This phase is sustainable only for a certain period before exhaustion sets in if the stressor remains unresolved.

Changes in the Individual During Stress

Exhaustion Phase:

Breakdown of Effort: Prolonged stress overwhelms the individual, leading to behavioral changes and emotional strain.

Depleted Resources: The body's physical and mental energy reserves are exhausted.

Persistent Stressors: The inability to adapt or eliminate stressors keeps the individual in a state of stress.

Increased Vulnerability: Long-term stress weakens resistance to other stressors, leading to further emotional and physical issues.

Critical Phase: Without intervention, this stage can result in severe health consequences and diminished coping capacity.

Physical Symptoms

Headache	Irregular sleep	Back pains	Jaw twitching or teeth grinding	Constipation, diarrhea and colitis
Debris	Muscle pains	Indigestion	Ulcer	Hypertension
Heart attack	Excessive sweating	Change in appetite	Fatigue or loss of energy	Increase in accidents

ERGONOMICS

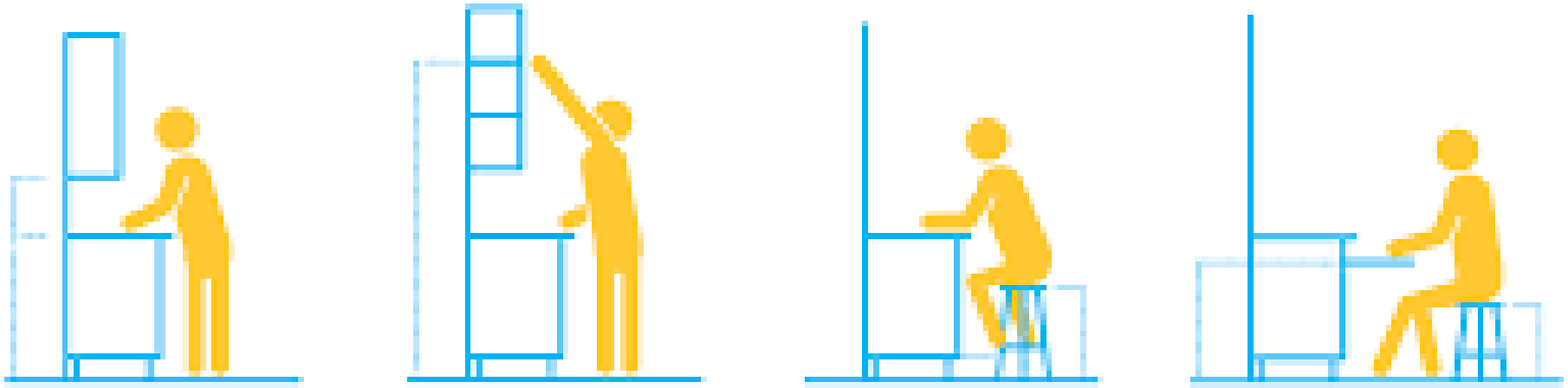
Ergonomic hazards

Philosophy of Ergonomics

Fit The Task To The Person

What is Ergonomics?

- Ergonomics is essentially the adaptation of the workplace to the employee.
- It is a branch of science which examines the interaction of human, equipment and environmental conditions and works to reduce and prevent physical and psychosocial problems arising from this interaction.



Ergonomics

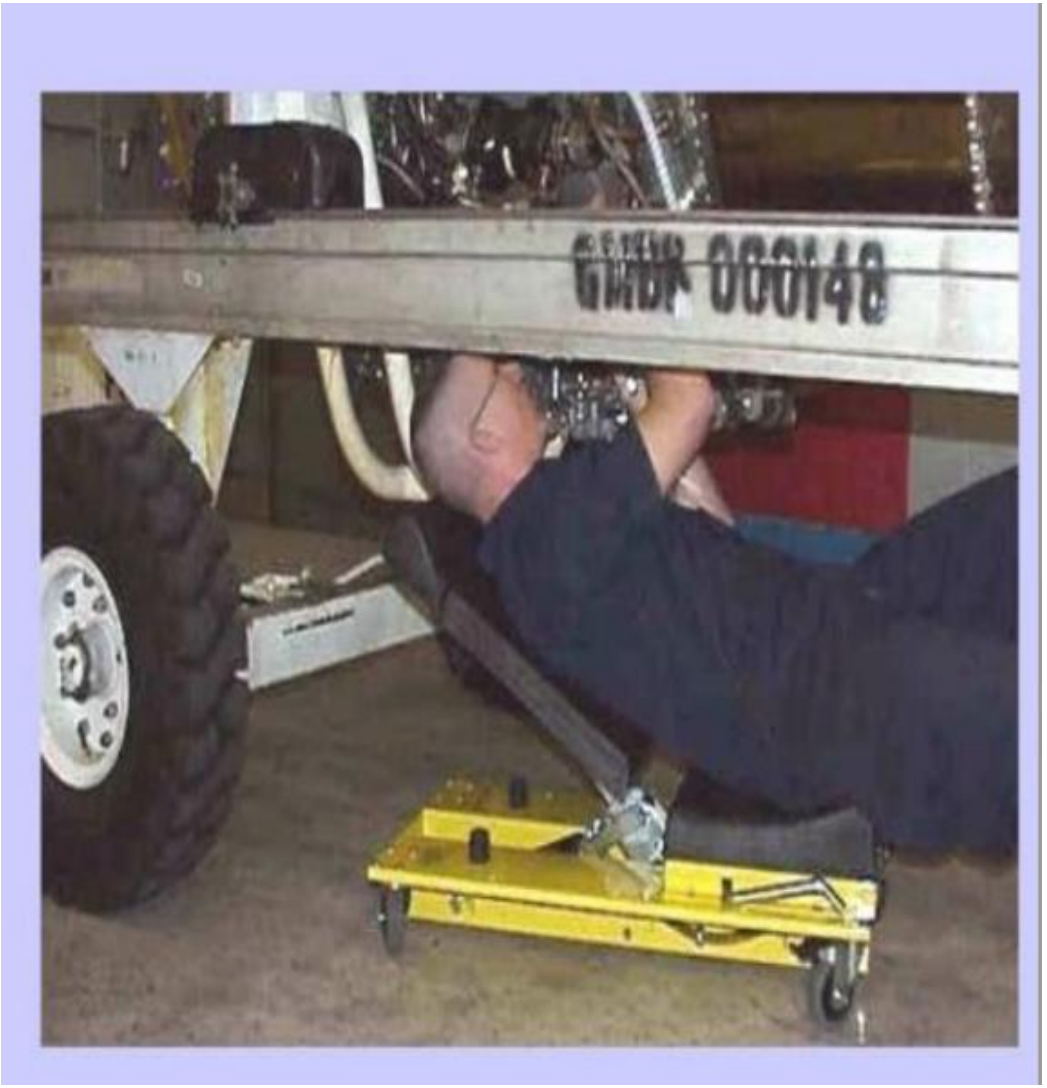
- Ergonomics is a combination of **ergon**, which means **work** in Greek, and **nomos**, which means **principles and laws**, and it means business science.

- Ergonomics is not a new science, although it has recently been used as a vocabulary. The Word was first used in 1857

Ergonomics

- It covers a wide range of scientific subjects as it is concerned with the improvement of working conditions that adversely affect workers in the workplace.
  - such as noise, lighting, heat,
  - vibration, working area design, hand tools design,
  - machine design, chair design, shoe design and
  - work design
  - In addition; working hours, shifts, break times and the food regime are also the main issues of ergonomics.
- The more compliance, the more safety and employee effectiveness.

Ergonomics



Ergonomics

BEFORE

AFTER

Ergonomics

Ergonomics

BEFORE

AFTER

Ergonomics

BEFORE

AFTER

Ergonomics

It is the science that compensate workload and capacity to work and aim both protect health of employees and increase the production.

It purpose importance to adaptation of person-machine-environment

Furnishing
Hand Tools
Equipment
Necessities
Media

COMPLIANCE

Anthropometric
Values
Skills
Limitations
Expectations

Ergonomics

What is anthropometry?

- Anthropometry is a science that deals with the determination and application of human body measurements. Engineering Anthropometry, on the other hand, is one of the most important topics of ergonomics, and it examines human measurements by evaluating them in terms of engineering.

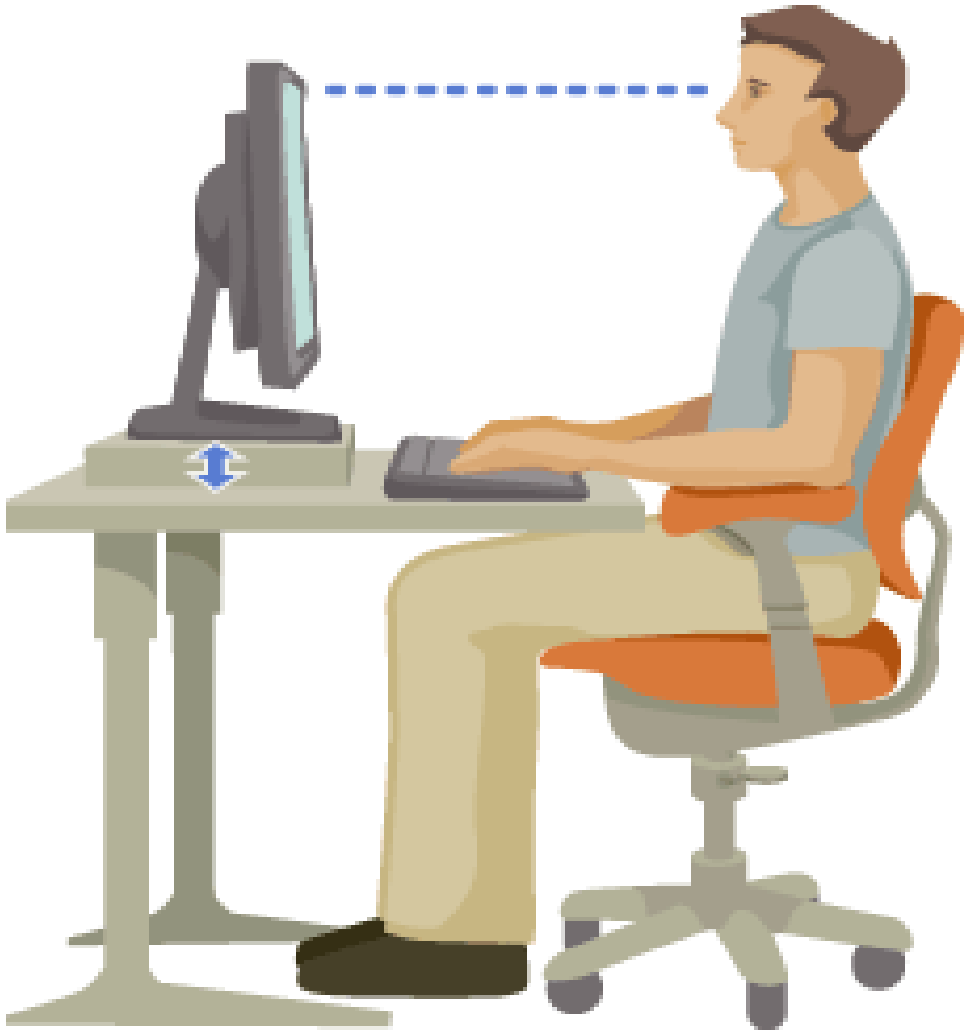
What does the arrangement of the workplace mean in anthropometric terms?

- It is the use of the science of Anthropometry to make (intersection for human-environment) designs of all kinds of tools and equipment by considering the size differences of users (which vary according to their age and gender).

Ergonomics

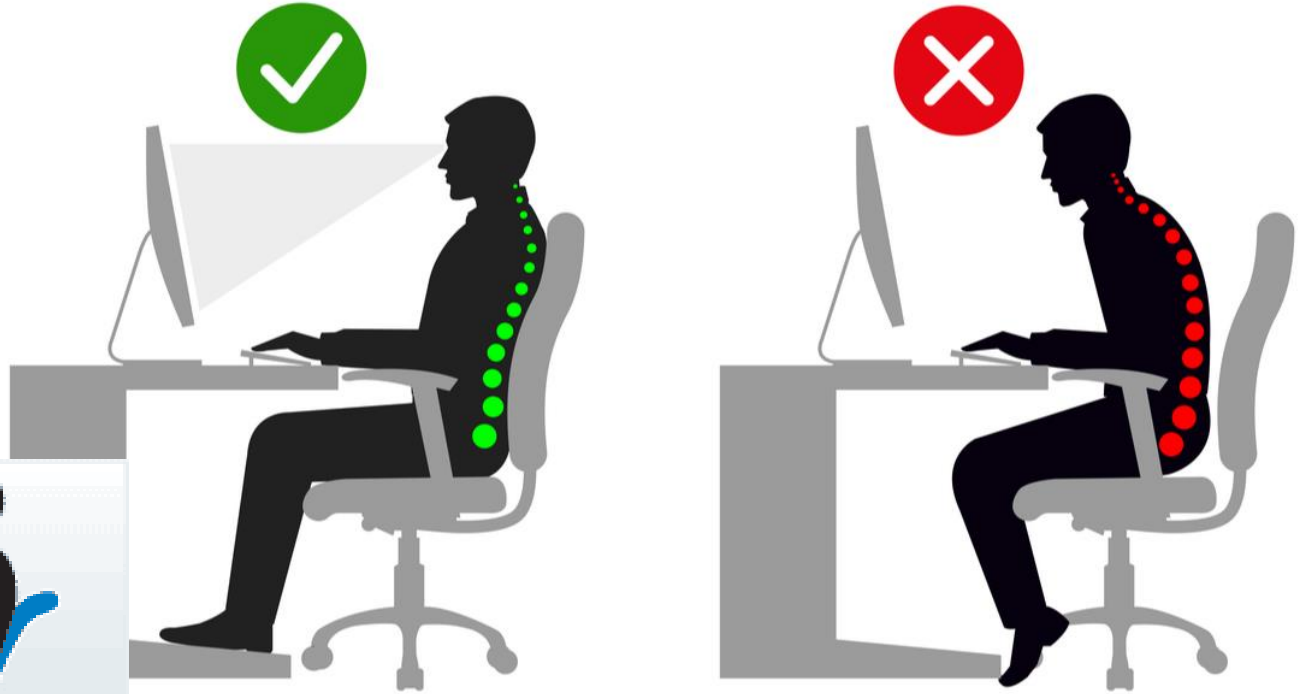
How far from the screen should you sit?

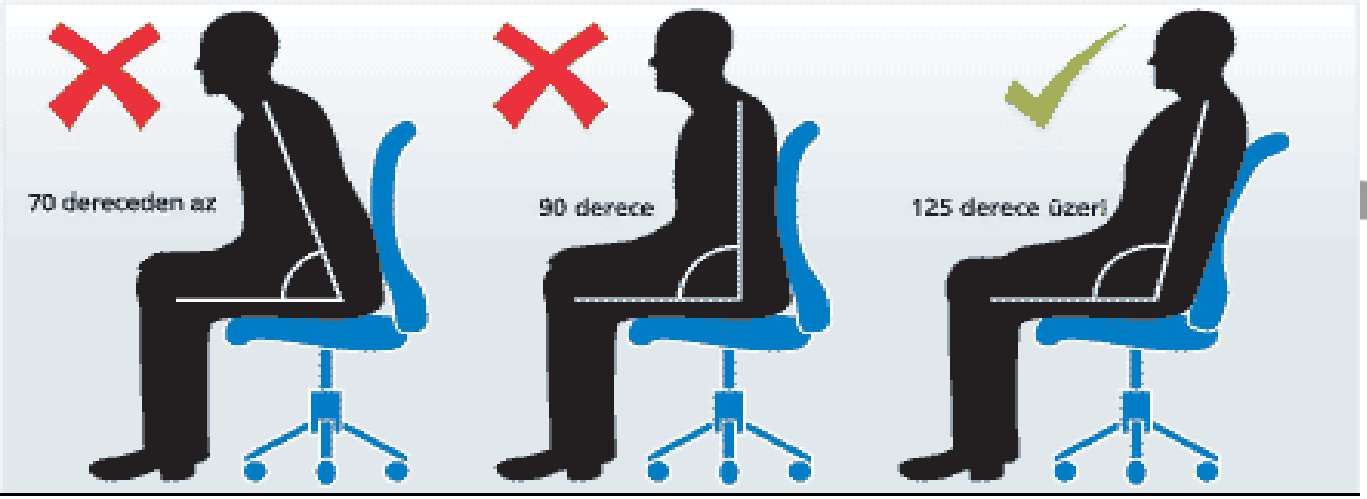
- Average, the eye-screen distance should be at least an arm distance of **(60 - 70 cm)**.



Ergonomics

POOR POSTURE MEANS MUSCLES WORKING MORE





Dry Eyes

- Under normal conditions, our eyes blink **15** times per minute, moisturizing the corneal surface, increasing its lubricity and washing the eye with its own acidic antiseptic liquid.

Dry Eyes

- The number of blinks is reduced by up to a third when looking at monitors and similar screens. The eyelids are fully opened, as if looking into the distance. In this state, the moisture on the surface of the eye, which reduces the number of blinks by two thirds, evaporates easily.

- The number of blinks decreases by two thirds in front of the screen, up to about 5 times

- The corneal surface dries up and irritation and erosion begin due to the friction of the lids and its resistance to microbes is weakened

$$\frac{1}{3}$$

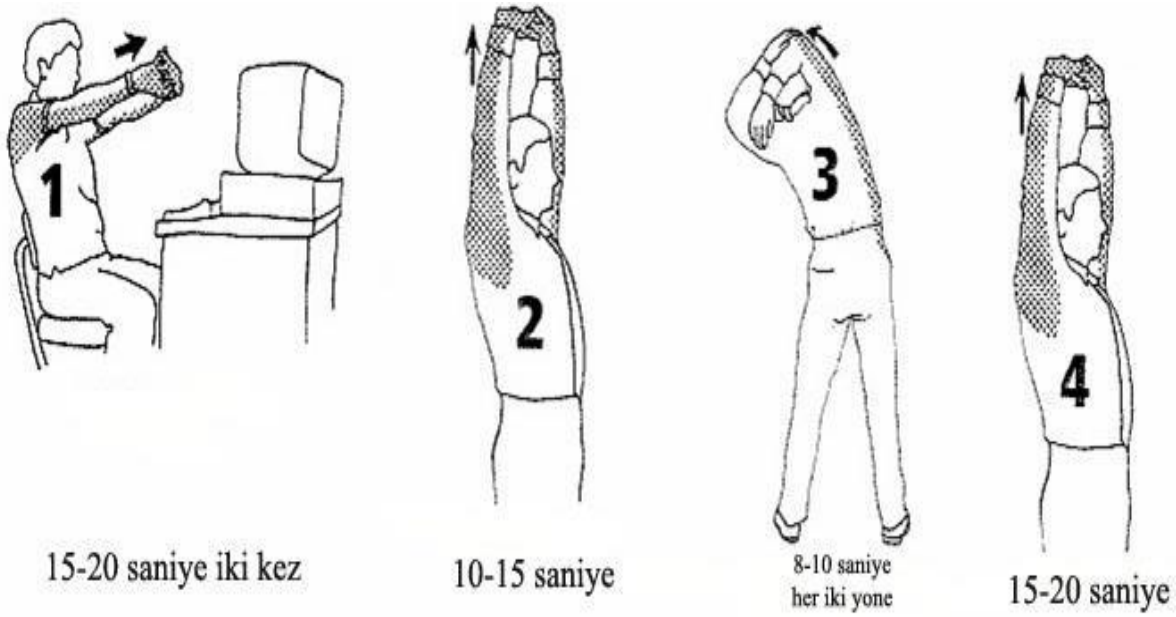
Dry Eyes

Prevention Tips:

- Take breaks following the 20-20-20 rule (look at something 20 feet away for 20 seconds every 20 minutes).
- Closing them for 10 to 15 seconds to rest.
- Blink more consciously while using screens.
- Use artificial tears if necessary to maintain moisture.

Office exercises

- Apply stretching exercises 3 - 4 minutes every hour



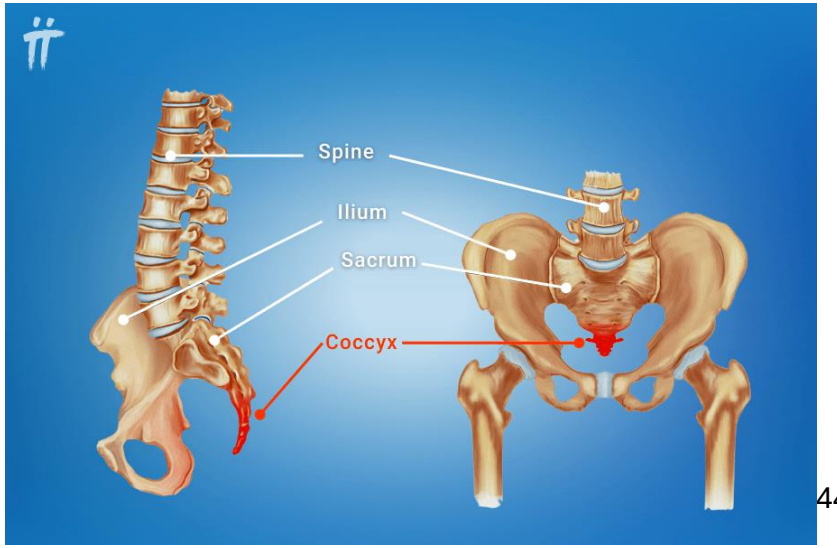
Office Exercises

- YOU SHOULD NOT FEEL PAIN OR STRAIN DURING THE EXERCISES, EXCEPT FOR THE FEELING OF STRETCHING.

Manual Lifting And Handling

Spine

- A bone called a spine is a part of the central nervous system, which extends from the neck to the coccyx in a structure and also includes a canal in the middle.



Back pain

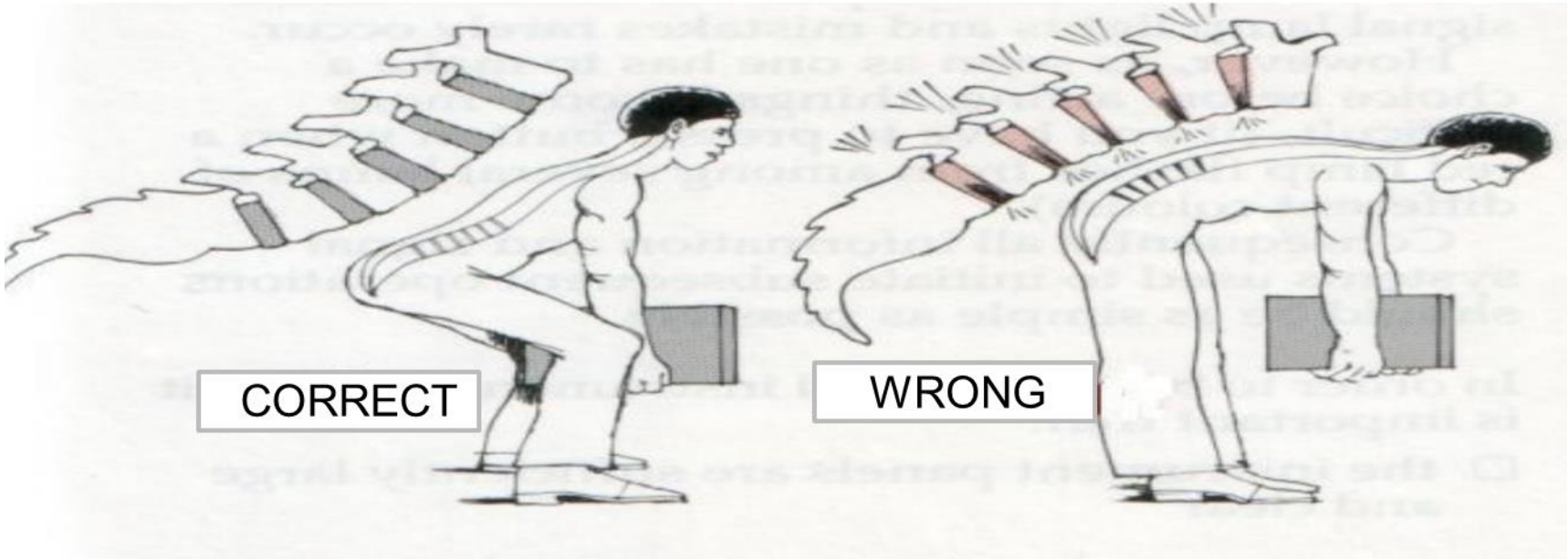
- Almost everyone suffers from low back pain at some point in their life.
- Back pain is the most common cause that restricts and disables physical functions in employees under the age of 45.
- Waist area is the most prone to injury.
- Acute pain is simple to treat, whereas chronic pain is challenging.
- The most common cause of low back pain is the disruption of soft tissue caused by overuse and improper body posture.

When to see a doctor?

- If there is pain and numbness in the leg
- If urine or fecal incontinence occurs
- If there is weakness in the foot or leg

Manual Lifting And Handling

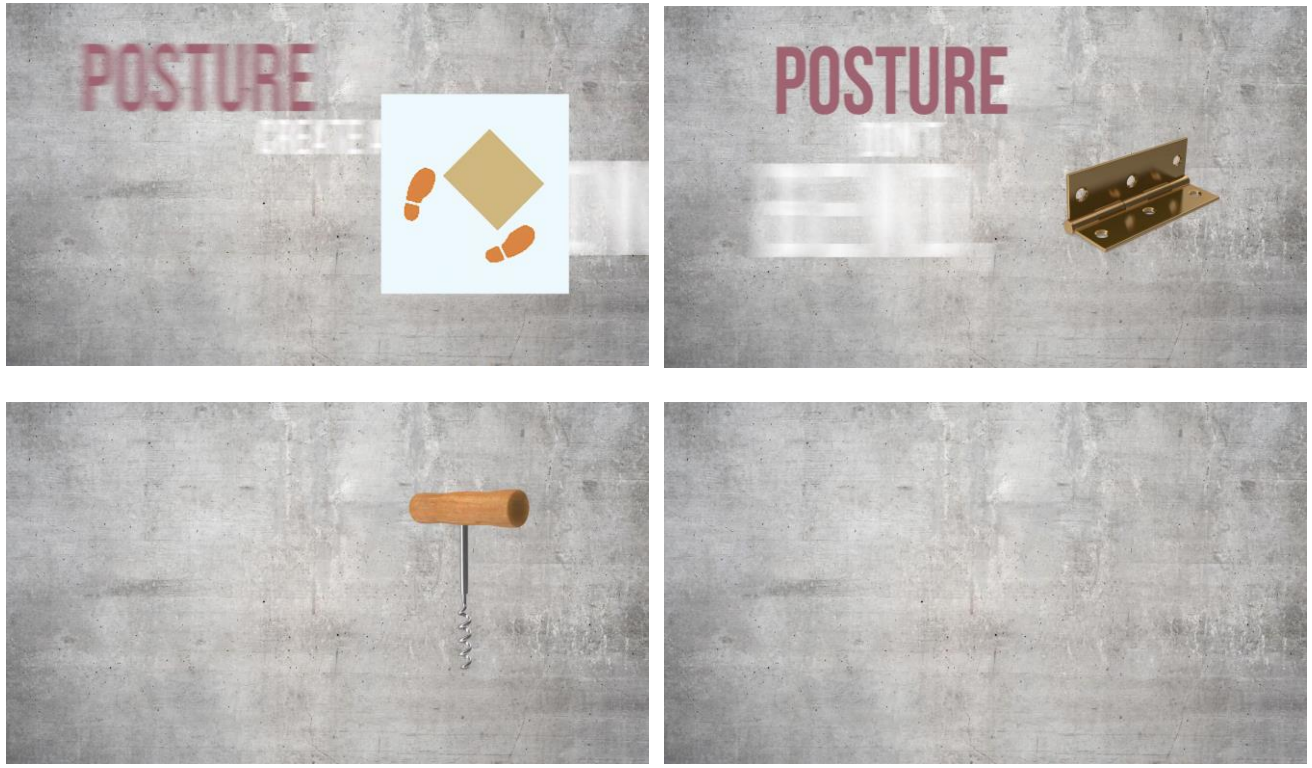
- The moves that strains the waist the most;
  - Leaning forward without bending the knees
  - Rotating the waist and torso with the hips fixed
  - Lifting heavy loads

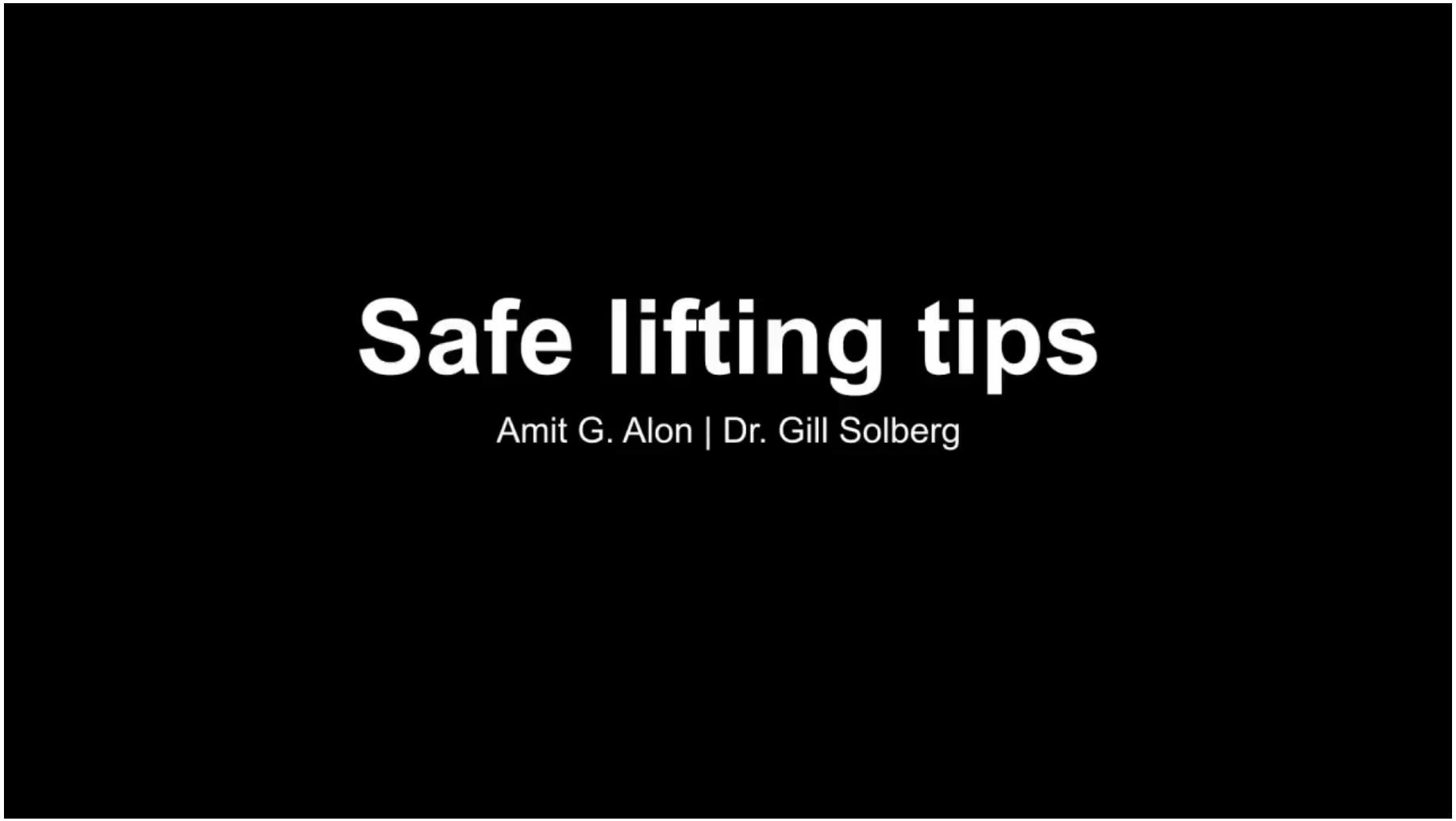


Manual Lifting And Handling

What should be considered when lifting a load?

- Lift the load by bending your knees
- Plan the lifting job
- Keep distance between feet
- Grasp the load without breaking your back



Manual Lifting And Handling

How To Lift Heavy Weight Safely